

[25/06/2026 10:39 AM] hssnd: Dr. Doc:
Well come to pharmacology class 😊👋

🌿 CHOLINERGIC AGENTS (PARASYMPATHOMIMETICS)

Definition

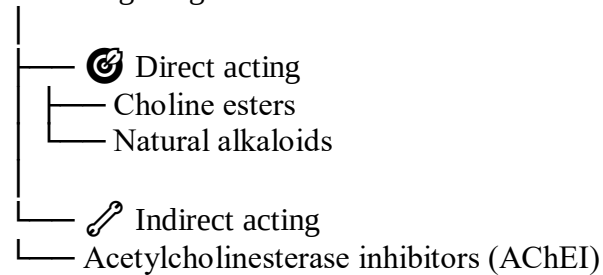
Cholinergic drugs are also called parasympathomimetics because they mimic the effects of acetylcholine (ACh) and increase parasympathetic activity.

Effects produced:

- ♥ ↓ Heart rate
- 👁 Miosis (pupil constriction)
- 👄 ↑ Salivation & secretions
- 🚶 ↑ GI motility
- 🚽 ↑ Bladder contraction
- ☐ Bronchoconstriction

Classification of Cholinergic Drugs

Cholinergic Agents



🎯 DIRECT-ACTING CHOLINERGIC AGENTS

Mechanism

They:

- ✅ Bind directly to receptors
- Muscarinic receptors
- Nicotinic receptors

A. Choline Esters 🌿

1. Acetylcholine (ACh) ☐

Characteristics:

- Ester of acetic acid + choline
- Highly hydrophilic 💧
- Rapidly hydrolyzed by AChE ✂
- Very short duration

Limitations:

- ✗ Poor absorption
- ✗ Cannot cross BBB

✗ Rapid destruction

2. Methacholine □

Modification:

+ β-methyl group

Effects:

✓ More muscarinic selective

✓ More resistant to AChE

✓ Longer action

3. Carbachol ●

Modification:

+ Carbamoyl group

Effects:

✓ Completely resistant to AChE

✗ No receptor selectivity

Acts on:

Muscarinic receptors

Nicotinic receptors

4. Bethanechol □

Modification:

+ β-methyl group

+ Carbamoyl group

Effects:

✓ Muscarinic selective

✓ Resistant to AChE

Drug of choice:

🚽 Urinary retention

🚫 GI atony

B. Naturally Occurring Alkaloids 🌿

Nicotine 🚬

Action:

✓ Nicotinic receptor agonist

Source:

🚬 Tobacco

Pilocarpine 👁💧

Action:

✓ Muscarinic receptor agonist

Uses:

👁 Glaucoma

💧 Xerostomia

Muscarine 🍄

Source:

🍄 Mushrooms

Action:

✓ Muscarinic receptor agonist

Effects:

🌀 Smooth muscle contraction

❤ ↓ Heart rate

🌀 ↑ Secretions

🔑 INDIRECT-ACTING CHOLINERGIC AGENTS

Mechanism

Normally:

ACh → AChE → Acetate + Choline ✂

Indirect drugs:

✕ Inhibit AChE

Result:

⬆ ACh concentration

Classification of AChE inhibitors

1. Reversible (Short acting) ⌚

Examples:

☐ Edrophonium

☐ Tacrine

Edrophonium ⚡

Uses:

🌀 Diagnosis of Myasthenia Gravis

Features:

✓ Rapid onset

✓ Rapid renal clearance

Tacrine ☐

Features:

- ✓ Lipid soluble
- ✓ Crosses BBB

Use:

☐ Alzheimer's disease

2. Carbamylating inhibitors (Intermediate acting) ⌚

Examples:

- ☐ Neostigmine
- ☐ Pyridostigmine
- ☐ Physostigmine

Duration:

⌚ 3–4 hours

Neostigmine + Pyridostigmine

Characteristics:

- ✗ Quaternary ammonium compounds
- ✗ Cannot cross BBB

Uses:

- 👉 Myasthenia gravis
- 👉 Reversal of neuromuscular blockade

Physostigmine

Characteristics:

- ✓ Crosses BBB

Use:

- 💀 Atropine poisoning

3. Phosphorylating inhibitors (Long acting) ⌚

Examples:

- ☐ Organophosphates
- ☐ Echothiophate
- ☐ Malathion

Effects:

- 🔒 Long-lasting AChE inhibition

Uses:

- 👁 Glaucoma
- 🐛 Insecticides

💀 ORGANOPHOSPHATE POISONING

In poisoning:

✗ AChE blocked



↑ ↑ ↑ Acetylcholine accumulates



Excess stimulation of:

🌿 Muscarinic receptors

⚡ Nicotinic receptors

☐ CNS receptors

Common causes:

🌿 Agricultural pesticides

🏠 Household insecticides

Examples:

☐ Malathion

☐ Parathion

Clinical Manifestations

Symptoms are divided into:

🌿 Muscarinic manifestations

⚡ Nicotinic manifestations

☐ CNS manifestations

1. Muscarinic manifestations 🌿

Memory trick:

SLUDE BAM

S → 🗑 Salivation

L → 😭 Lacrimation

U → 🚽 Urination

D → 🚽 Defecation

E → ☐ Emesis

B → ❤ Bradycardia + ☐ Bronchoconstriction

A → 😭 Abdominal cramps

M → 👁 Miosis (pinpoint pupils)

2. Nicotinic manifestations ⚡

Excess stimulation of neuromuscular junction:

Initially:

🌀 Fasciculations (muscle twitching)



🌀 Weakness



✗ Paralysis

This is called:

Depolarizing blockade

Serious effects:

☐ Respiratory muscle paralysis

♡ Cardiovascular collapse

3. CNS manifestations ☐

☐ Anxiety

☐ Confusion

☐ Restlessness

☐ Convulsions/seizures

☐ Respiratory depression

☐ Coma

Treatment  

[25/06/2026 10:39 AM] hssnd: Treatment

Step 1: Maintain vital signs 

Priority:

☐ Airway and breathing support

Because:

Respiratory failure is the major cause of death

May require:

Oxygen

Mechanical ventilation

Step 2: Decontamination 

Prevent further absorption:

 Remove contaminated clothes

☐ Wash skin thoroughly

Step 3: Atropine + Pralidoxime 

Atropine  ✗

Mechanism:

Blocks muscarinic receptors

Improves:

 Salivation

☐ Bronchoconstriction

♡ Bradycardia

⚠ Important:

Does NOT reverse muscle paralysis

Pralidoxime (2-PAM) 

Mechanism:

Reactivates acetylcholinesterase

Very important:

 Give EARLY

Aging ☐

After some time:

Organophosphate + enzyme



 Permanent bond forms

After aging:

✗ Pralidoxime becomes ineffective

Step 4: Benzodiazepines 

Examples:

Diazepam

Lorazepam

Use:

☐ Control seizures

Therapeutic Uses of Cholinomimetics

1.  Glaucoma

Disease

Progressive optic nerve damage associated with:

 Intraocular pressure (IOP > 21 mmHg)

Mechanism of cholinomimetics

 Miosis (pupil constriction)



 Increased aqueous humor drainage



 IOP

Drugs:

☐ Pilocarpine

☐ Physostigmine

☐ Echothiophate

2.  Myasthenia Gravis

Disease

Autoimmune destruction of nicotinic receptors at the neuromuscular junction.

Symptoms:

- 👉 Muscle weakness
- 👉 Fatigue with use

Mechanism of cholinomimetics

- 🔧 Inhibit AChE



- ↑ ACh in synaptic cleft



More stimulation of remaining receptors



- 👉 Improved muscle contraction

Drugs:

- ☐ Pyridostigmine
- ☐ Neostigmine

3. 🚽 Urinary Retention

Problem

Bladder cannot contract properly.

Mechanism

- ↑ Detrusor muscle contraction



- 🚽 Improves urination

Drug of choice:

- ☐ Bethanechol

4. 🍷 Gastrointestinal Disorders

Examples:

- 🍷 Gastric atony
- 🍷 Gastric paresis

Common in:

- ☐ Diabetes patients

Mechanism

- ↑ GI smooth muscle activity



- ↑ Gastric emptying

Drug:

- ☐ Bethanechol

5. Xerostomia (Dry Mouth)

Disease

Reduced salivary and lacrimal secretion.

Symptoms:

 Dry mouth

 Dry eyes

Mechanism

 Salivary gland secretion

 Lacrimal gland secretion

Drug:

☐ Pilocarpine

6. Alzheimer's disease

Disease

Progressive loss of:

☐ Memory

☐ Cognitive function

Mechanism

Drugs cross BBB and:

 inhibit AChE



 CNS acetylcholine

Drugs:

☐ Donepezil

☐ Rivastigmine

☐ Tacrine

☐ Galantamine

7. Reversal of Neuromuscular Blockade

Mechanism

Increase ACh at neuromuscular junction



Competes with muscle relaxants

Drugs:

☐ Neostigmine

Use:

Recovery after surgery

8. Antimuscarinic (Atropine) Poisoning

Mechanism

Increase ACh in CNS and periphery

Drug:

☐ Physostigmine

Reason:

☒ Crosses BBB

Contraindications ⊗

☐ Asthma → bronchoconstriction

☐ Hypotension → may worsen BP drop

♥ Heart failure

🍷 Peptic ulcer → ↑ acid secretion

🚫 GI obstruction

🚫 Urinary tract obstruction

Anticholinergics / Antimuscarinics

✗ Block the action of acetylcholine (ACh) at muscarinic receptors

Because parasympathetic actions are blocked, the effects become almost the opposite of parasympathetic activity.

Classification of Antimuscarinics

1. Tertiary amines □

These are lipid soluble

☒ Cross blood-brain barrier (BBB)

☒ Can cause CNS effects

Examples:

☐ Atropine

☐ Homatropine

☐ Hyoscine (Scopolamine)

☐ Pirenzepine

☐ Benztropine / Trihexyphenidyl

☐ Tropicamide

☐ Cyclopentolate

2. Quaternary amines ⊗ □

Characteristics:

✓ Highly polar

✓ Cannot cross BBB

Examples:

☐ Ipratropium bromide

☐ Propantheline bromide

Therapeutic Uses of Antimuscarinics 🍯

1. Parkinson disease □

Mechanism:

In Parkinson disease:

☐ ↓ Dopamine

causes relative:

☐ ↑ Cholinergic activity in the striatum

Antimuscarinics reduce this excess cholinergic activity.

Drug:

☐ Benztropine

☐ Trihexyphenidyl

[25/06/2026 10:39 AM] hssnd: Main treatment still focuses on replacing dopamine:

☐ Levodopa

Antimuscarinics are usually:

Mild cases

Combination therapy

2. Motion sickness

Drug:

☐ Scopolamine (Hyoscine)

Mechanism:

Blocks cholinergic pathways involved in motion-induced vomiting.

Use:

 Prevention of motion sickness

3. Ophthalmology

Purpose:

 Dilate pupil (mydriasis)

Use:

Retinal examination (ophthalmoscopy)

Examples:

☐ Tropicamide

☐ Cyclopentolate

☐ Atropine

Effects:

 Pupil dilation

 Cycloplegia (loss of accommodation)

4. Respiratory disorders ☐

Atropine

Used before surgery:

Purpose:

Reduce airway secretions

Especially with inhalation anesthetics

Ipratropium bromide ☐

Uses:

- ☐ Bronchial asthma
- ☐ COPD

Mechanism:

Blocks muscarinic receptors in airways



- ☐ Bronchodilation



- ⇒ Improved airflow

Why Ipratropium is different ☆

- ☒ Given by inhalation
- ☒ Quaternary ammonium compound
- ☒ Does not cross BBB
- ☒ Fewer CNS side effects

⚠ Adverse Effects

(Occurs because antimuscarinics block parasympathetic activity)

👁 Blurred vision → mydriasis + cycloplegia

☼ Photophobia → dilated pupils allow more light in

👄 Dry mouth (xerostomia) → ↓ salivary secretion

🚽 Urinary retention → ↓ bladder contraction

🚽 Constipation → ↓ GI motility

👁 ☐ ↑ Increased intraocular pressure (IOP) → may worsen glaucoma

💓 Tachycardia → blockade of vagal effect on heart

☐ CNS effects (high doses) → restlessness, confusion, hallucinations

⊘ Contraindications

👁 Glaucoma → may increase IOP

🚽 Benign Prostatic Hyperplasia → worsens urinary retention

🚽 Bladder atony → further reduces bladder emptying

🚫 GI atony/obstruction → worsens constipation or ileus

🦋 Myasthenia Gravis → may worsen muscle weakness

💓 Tachydysrhythmias → may increase heart rate further

🚫 Peptic ulcer disease (relative caution)

💊 Effects of Atropine

☐ Atropine = Antimuscarinic (Parasympatholytic)

Mechanism:

✗ Blocks muscarinic receptors

Result:

🚫 Blocks parasympathetic (“rest and digest”) actions

Effects on Different Organs

1. Skin

Effects:

✗ Decreased sweating



🔥 Increased body temperature



🤒 Fever (especially in children)

Sweat glands normally receive cholinergic stimulation.

Blocking them:

✗ No sweat



Body cannot lose heat efficiently

2. CNS ☐

Effects:

☐ Restlessness

High doses may cause:

☐ Confusion

☐ Agitation

☐ Hallucinations

Some antimuscarinics (like benztropine) also have:

 Antiparkinsonian effects

3. Salivary glands

Effects:

✗ Decreased salivary secretion

Result:

 Xerostomia (dry mouth)

4. Eye

Effects:

 Mydriasis = Pupil dilation

 Cycloplegia= Loss of accommodation for near vision

 Decreased tears

Effects:

✗ Dry eyes

5. Heart

Effects:

 Heart rate

Reason:

Normally parasympathetic activity slows the heart.

Atropine blocks that action.

Result:

 Stimulation (tachycardia)

6. Bronchi

Effects:

 Bronchodilation

✗ Decreased bronchial secretions

Uses:

 Preoperative medication

7. Gastrointestinal tract

Effects:

✗ Decreased HCl secretion

✗ Decreased GI motility

Results:

 Slow digestion

 Constipation

8. Urinary tract 🚽

Effects:

✗ Decreased bladder contraction

Result:

🚽 Urinary retention

Difficulty emptying bladder

[25/06/2026 11:20 AM] hssnd: Dr. Doc:

🔗 ADRENERGIC DRUGS (SYMPATHOMIMETICS)

Adrenergic drugs (Sympathomimetics) are drugs that mimic the actions of the sympathetic nervous system by:

✓ Directly stimulating adrenergic receptors (α or β)

OR

✓ Increasing norepinephrine (NE) at adrenergic synapses

□ Classification of Adrenergic Drugs

1 Direct-Acting Adrenergic Agonists

Act directly on adrenergic receptors.

Catecholamines

◆ Adrenaline (Epinephrine)

◆ Noradrenaline (Norepinephrine)

◆ Dopamine

◆ Isoproterenol

◆ Dobutamine

Non-catecholamines

◆ Phenylephrine

◆ Clonidine

◆ Methyldopa

2 Indirect-Acting Adrenergic Agonists

Do not stimulate receptors directly.

Increase NE or DA in synapse.

✓ Tyramine 🎲

Releases stored NE

✓ Amphetamine ⚡

Releases NE and DA

Blocks reuptake

✓ Cocaine 🌸

Blocks reuptake of NE and DA

3 Mixed-Acting Adrenergic Agonists

✓ Ephedrine 

Acts:

Directly on α and β receptors

Indirectly by releasing NE

□ Catecholamines vs Non-Catecholamines

Catecholamines

Characteristics

⚡ Rapid onset

⌚ Short duration

 Must be given parenterally

✗ Ineffective orally

□ Poor CNS penetration

Potency

α Receptors

◆ NE > E > Isoproterenol

β Receptors

◆ Isoproterenol > E > NE

Non-Catecholamines

Characteristics

✓ Oral administration

⌚ Longer duration

□ Better CNS penetration

● ALPHA (α) RECEPTORS

Two Main Types

✓ α_1 Receptors

✓ α_2 Receptors

α_1 ADRENORECEPTORS

 Location

▲ □ Blood Vessels

Especially:

Skin

Viscera

▲  Iris Radial Muscle

▲  GIT Sphincters

▲  Internal Urinary Bladder Sphincter

Effects of α_1 Stimulation

☐ 1. Vasoconstriction

α_1 stimulation causes contraction of vascular smooth muscle.

Result:

☐ ↑ Peripheral vascular resistance (PVR)

☐ ↑ Blood pressure

2. Nasal Decongestion

α_1 stimulation constricts nasal blood vessels.

Result:

☐ ↓ Edema

☐ ↓ Congestion

☐ ↑ Airflow

3. Mydriasis

α_1 stimulates radial muscle of iris.

Result:

 Pupil dilates

4. GIT Effects

α_1 stimulation causes contraction of:

Intestinal sphincters

Gastrointestinal sphincters

Result:

☐ ↓ GI motility

5. Urinary Effects

α_{1A} receptors stimulate internal sphincter.

Result:

☐ Closure of bladder outlet

☐ Urinary retention

α_1 SELECTIVE AGONISTS

Phenylephrine

Type

✓ Direct-acting α_1 agonist

✓ Non-catecholamine

Pharmacokinetics

✓ Not metabolized by COMT rapidly.

Therefore:

✓ Longer duration

✓ Oral activity possible

Uses of Phenylephrine

👁 Mydriatic

Fundus examination.

α_1 → radial muscle contraction
→ pupil dilation

👃 Nasal Decongestant

α_1 vasoconstriction
→ reduced swelling

☐ Hypotension

α_1 vasoconstriction
→ increased BP

Xylometazoline

Oxymetazoline

✓ Used as topical nasal decongestants.

Mechanism

α receptor stimulation
→ vasoconstriction
→ relief of congestion

Adverse Effects of α_1 Agonists

☑ Hypertension

Excess vasoconstriction

💓 Reflex Bradycardia

BP rises

↓

Baroreceptors activated

↓

Vagus nerve stimulated

↓

HR decreases

👁 Glaucoma Risk

May increase intraocular pressure.



α_2 ADRENORECEPTORS

📍 Location

☐ CNS Cardiovascular Control Center

🔗 Presynaptic Adrenergic Nerve Terminals

☐ Platelets

⚙️ Effects of α_2 Stimulation

🔗 1. Decreased Norepinephrine Release

Presynaptic α_2 receptors act as negative feedback receptors.

When stimulated:

⬇️ NE release

☐ 2. Reduced Sympathetic Outflow

Central α_2 stimulation suppresses sympathetic activity.

Result:

⬇️ PVR

⬇️ BP

⬇️ HR

☐ 3. Increased Platelet Aggregation

α_2 stimulation enhances platelet clumping.

💊 α_2 Selective Agonists

Clonidine

Although it is an adrenergic agonist,

➡️ It acts as a sympatholytic drug.

Stimulates central α_2 receptors

↓

Less NE release

↓

Less sympathetic activity

↓

Lower BP

Uses of Clonidine

💎 Hypertension

Reduces:

Sympathetic tone

PVR

Blood pressure

💎 Migraine Prophylaxis

Low dose

50–100 mcg/day

Adverse Effects of Clonidine



Sedation

Because it suppresses CNS sympathetic activity.

[25/06/2026 11:20 AM] hssnd: Dry Mouth



Bradycardia

Less sympathetic stimulation to heart.



Depression

May occur with prolonged use.



Sexual Dysfunction

Due to decreased sympathetic activity.

Withdrawal Syndrome

Abrupt discontinuation



Sudden sympathetic overactivity



Rebound hypertension

Clonidine Overdose

Large doses stimulate peripheral α_1 receptors.

Result:



Severe hypertension

because vasoconstriction predominates.

Methyldopa

Mechanism

Converted in brain to:

α -methylnorepinephrine



Stimulates central α_2 receptors



Decreases sympathetic outflow



Lowers BP

Major Use



Hypertension in Pregnancy

Safest and most preferred antihypertensive during pregnancy.



β_1 & β_2 ADRENORECEPTORS



β_1 ADRENORECEPTORS



Location

♥ Heart

Found in:

SA node

AV node

Conducting system

Myocardium

□ Kidney

Found in:

Juxtaglomerular (JG) cells

⚙ Effects of β_1 Stimulation

♥ 1. SA Node

Positive Chronotropic Effect

➡ Increased Heart Rate

β_1 increases automaticity of SA node.

Result:

♥ Tachycardia

♥ 2. Myocardium

Positive Inotropic Effect

➡ Increased Force of Contraction

β_1 increases intracellular calcium.

Result:

♥ Stronger heartbeat

♥ 3. Conducting System

Positive Dromotropic Effect

➡ Increased Conduction Velocity

Electrical impulses travel faster.

Result:

⚡ Faster AV conduction

□ 4. Kidney

Increased Renin Release

β_1 stimulates JG cells.

Result:

Renin ↑

↓

Angiotensin II ↑

↓

Aldosterone ↑

↓

BP ↑

β_1 SELECTIVE AGONIST

Dobutamine

Mechanism

Stimulates β_1 receptors.

Produces:

♥ Strong increase in contractility
with

⬆ Minimal increase in heart rate

Clinical Uses

🚑 Cardiogenic Shock

Heart is failing to pump blood.

Dobutamine:

⬆ Cardiac output

⬆ Tissue perfusion

♥ Acute Heart Failure

Especially after:

✓ Myocardial infarction

✓ Cardiac surgery

Adverse Effects

♥ Tachycardia

⚡ Arrhythmias

☑ Increased myocardial oxygen demand

May worsen ischemia.

☐ β_2 ADRENORECEPTORS

📍 Location

☐ Bronchial Smooth Muscle

☐ Blood Vessels Supplying Skeletal Muscle

👶 Uterus

🍲 GIT Wall

🚽 Urinary Bladder Wall

🦘 Liver

☐ Pancreas

👁 Ciliary Muscle

👁 Ciliary Epithelium

⚙ Effects of β_2 Stimulation

☐ 1. Bronchodilation

β_2 relaxes bronchial smooth muscle.

Clinical Importance

Most important β_2 effect.

Used in:

- ☒ Asthma
- ☒ COPD

☐ 2. Vasodilation

Occurs mainly in vessels supplying skeletal muscles.

Result:

- ☐ ↓ Peripheral resistance
- ☐ ↓ Diastolic BP

3. GIT Relaxation

β_2 relaxes intestinal wall.

Result:

- ☐ ↓ Peristalsis

4. Bladder Relaxation

β_2 relaxes bladder wall.

Result:

- ☐ ↑ Urine storage

5. Uterine Relaxation

β_2 relaxes uterine smooth muscle.

Result:

Delayed contractions

Clinical Use

Prevent premature labor.

6. Liver Effects

β_2 stimulation causes:

- ◆ Glycogenolysis
- ◆ Gluconeogenesis

Result:

- ☒ Blood sugar increases

☐ 7. Pancreatic Effects

β_2 causes:

- ☐ ↑ Insulin secretion

Liver glucose production is much greater than Insulin increase

Final result:

- ☒ Hyperglycemia

👁 Eye Effects

✓ Ciliary Muscle Relaxation

Result:

👁 Accommodation for far vision

✓ Ciliary Epithelium

β_2 stimulation:

⬆ Aqueous humor production

? Can β_2 Agonists Be Used For Glaucoma?

✗ No

Because they increase aqueous humor production.

May worsen glaucoma.

💊 β_2 SELECTIVE AGONISTS

Short-Acting β_2 Agonists (SABA)

✓ Salbutamol

✓ Terbutaline

✓ Metaproterenol

Uses

Acute Asthma Attack

Acute Bronchospasm

Rescue Therapy

Long-Acting β_2 Agonists (LABA)

Salmeterol

Formoterol

Uses

🌙 Nocturnal Asthma

Long-Term Asthma Control

NOT for acute attacks.

💊 Ritodrine

Ritodrine

Use

Prevent premature labor.

β_2 stimulation

↓

Uterine relaxation

↓

Delayed labor

Adverse Effects of β_2 Agonists

✋ Skeletal Muscle Tremor

β_2 receptors in skeletal muscle.

♥ Tachycardia

Occurs because:

1. β_1 stimulation (high doses)

OR

2. Reflex response to β_2 vasodilation

⚡ Arrhythmias

Excess cardiac stimulation.

😬 Anxiety

CNS stimulation.

☑ Hyperglycemia

β_2 stimulates:

Glycogenolysis

Gluconeogenesis

[25/06/2026 11:20 AM] hssnd: ⚡ INDIRECTLY ACTING ADRENOMIMETICS

Indirectly acting adrenomimetics do NOT stimulate α or β receptors directly.

Instead, they increase the concentration of:

◆ Norepinephrine (NE) ♥

◆ Dopamine (DA) □

at adrenergic synapses.

Result:

→ Increased sympathetic activity

→ Adrenergic receptors are stimulated indirectly

Direct Agonists

Stimulate receptors themselves.

Example:

✓ Phenylephrine → α_1

✓ Dobutamine → β_1

Indirect Agonists

Do NOT stimulate receptors.

Instead:

● Release NE

OR

⊖ Block NE reuptake

Result:

More NE available to stimulate receptors.

📖 Classification

📁 Releasing Agents

Cause release of stored norepinephrine.

 Tyramine

 Amphetamine

2 Reuptake Inhibitors

Prevent neurotransmitter reuptake.

 Cocaine

 TYRAMINE

Source

✓ Naturally occurring by-product of tyrosine metabolism.

Found in:

✓ Aged cheese 

✓ Fish 

✓ Fermented foods

Mechanism

Tyramine resembles norepinephrine.

Therefore it enters adrenergic nerve terminals.

Inside the nerve terminal:

Displaces stored NE from vesicles

↓

NE released into synapse

↓

Sympathetic stimulation

Normally tyramine is rapidly metabolized by:

✓ MAO in gut

✓ MAO in liver

Therefore very little reaches circulation.

 Tyramine + MAOI = Cheese Reaction

When patient takes MAO inhibitors:

✗ Tyramine not destroyed

↓

Large amount enters blood

↓

Massive NE release

↓

Severe vasoconstriction

↓

 Hypertensive crisis

 Cheese + MAOI = Hypertensive Crisis

 AMPHETAMINE

Type

- ✓ Indirectly acting sympathomimetic
- ✓ Non-catecholamine

Mechanism

- ✓ Releases Norepinephrine
- Enters adrenergic nerve terminal
- ↓
- Displaces NE from storage vesicles
- ↓
- NE released

- ✓ Releases Dopamine
- Increases DA in CNS

- ✓ Blocks Reuptake
- Prevents reuptake of NE
- Result:
- More NE remains in synapse.

Receptor Effects

Because NE increases:

α_1 stimulation

- ☐ Vasoconstriction
- ☒ Increased BP

β_1 stimulation

- ♥ Increased HR
- ♥ Increased contractility

Minimal β_2 effect

Because NE has weak β_2 activity.

Pharmacokinetics

- ✓ Non-catecholamine
- Therefore:
- ☒ Oral administration
 - ☒ Longer duration

- ✓ Lipid Soluble
- Crosses BBB easily.
- Result:
- Strong CNS effects.

CNS Effects

- ☐ Alertness
- ☐ Increased concentration
- ☐ Reduced fatigue
- 😊 Euphoria

- 🌙 Insomnia
- 😓 Restlessness

Clinical Uses

- 🎯 ADHD

Increases NE and DA in brain.

Improves:

- ◆ Attention
- ◆ Focus

- 😴 Narcolepsy

Promotes wakefulness.

- 🏠 Appetite Suppression

- ◆ Suppresses hunger center.
- ◆ Used in severe obesity.

Adverse Effects

- ❤️ Hypertension
 α_1 stimulation

- ❤️ Tachycardia
 β_1 stimulation

- 🌙 Insomnia
CNS stimulation

- 🏠 Tolerance
Need higher doses.

- 🏠 Dependence
Body adapts to drug.

- 🏠 Addiction
Dopamine reward pathway stimulation.

- 🏠 Psychosis
Excess dopamine
↓
Hallucinations
↓
Paranoia
↓
Delusions

- 🌸 COCAINE

Type

- ✓ Indirect sympathomimetic
- ✓ Local anesthetic

Mechanism

- ◆ Blocks NE Reuptake

CNS Action

Blocks dopamine reuptake.

Result:

- Massive dopamine accumulation

↓

- 😊 Intense euphoria

Cocaine is highly addictive

Because dopamine accumulates in:

- Brain pleasure centers

Local Anesthetic Action

Blocks sodium channels.

Result:

- 🪡 Loss of sensation

Effects

α_1

- Vasoconstriction
- ☑ Hypertension

β_1

- ♥ Tachycardia
- ♥ Increased cardiac work

CNS

- 😊 Euphoria
- ⚡ Alertness
- 🧠 Addiction

Adverse Effects

- ♥ Hypertension
- ⚡ Arrhythmias
- ♥ Myocardial infarction
- Seizures
- Psychosis
- 🧠 Dependence

- 🎲 Tyramine

- ◆ Found in cheese
- ◆ Releases NE
- ◆ MAOI + Tyramine = Hypertensive crisis

⚡ Amphetamine

- ◆ Releases NE & DA
- ◆ ADHD
- ◆ Narcolepsy
- ◆ Appetite suppression
- ◆ Psychosis

Cocaine

- ◆ Blocks NE reuptake
- ◆ Blocks DA reuptake
- ◆ Local anesthetic
- ◆ Intense euphoria
- ◆ Highly addictive

□ Hypertension (HTN)

Hypertension (HTN) is a persistent elevation of arterial blood pressure (BP) above normal levels.

□ Blood Pressure (BP) = Pressure exerted by blood on the walls of arteries.

It is measured in millimeters of mercury (mmHg).

[25/06/2026 11:20 AM] hssnd: Example:

120/80 mmHg

↑ ↑

SBP DBP

□ Components of Blood Pressure

1 Systolic Blood Pressure (SBP)

SBP is the maximum arterial pressure reached during ventricular contraction (systole). ♥

Heart contracts

↓

Blood ejected into aorta

↓

Peak arterial pressure

↓

SBP

Example:

□ 120 mmHg

2 Diastolic Blood Pressure (DBP)

DBP is the minimum arterial pressure during ventricular relaxation (diastole).

Heart relaxes

↓

Ventricles fill with blood

↓

Lowest arterial pressure

↓

DBP

Example:

☐ 80 mmHg

 Types of Hypertension

Hypertension is classified into:

1 Primary (Essential) Hypertension

Definition

Hypertension with no identifiable cause.

Frequency

 More than 90% of hypertensive patients have primary hypertension.

Hypertension cases

↓

>90%

↓

Essential HTN

Characteristics

☒ Cause unknown

☒ Multifactorial

☒ Cannot be cured

☒ Can be controlled with lifestyle changes and drugs

Contributing Factors

☐ Genetics

 Obesity

☐ Excess salt intake

 Stress

 Smoking

 Physical inactivity

2 Secondary Hypertension

Definition

Hypertension resulting from a specific identifiable cause.

Frequency

 Less than 10% of hypertension cases.

Hypertension cases

↓

<10%

↓

Secondary HTN

Important Feature

 If the cause is corrected, hypertension may be cured.

 Causes of Secondary Hypertension

1 Chronic Kidney Disease (CKD) ☐

Most common cause.

Damaged kidneys:

Kidney disease



↑ Renin release



↑ Angiotensin II



↑ BP

2 Renovascular Disease

Narrowing of renal arteries.

Renal artery stenosis



Kidney senses low blood flow



↑ Renin secretion



↑ BP

3 Primary Aldosteronism

Excess aldosterone production.

Usually from:

☐ Adrenal adenoma

Effects:

↑ Aldosterone



↑ Na⁺ retention



↑ Water retention



↑ Blood volume



↑ BP

4 Pheochromocytoma



Pheochromocytoma

Produces:

☐ Norepinephrine

☐ Epinephrine

Result:

↑ Catecholamines



Vasoconstriction



↑ BP

Classic triad:



Headache



Palpitations

🌀 Sweating

5 📖 Cushing Syndrome

Excess cortisol production.

↑ Cortisol

↓

↑ Sensitivity to catecholamines

↓

↑ BP

6 📖 Drugs Causing Secondary HTN 📖

Amphetamines

📖 Amphetamine

Increase norepinephrine release.

Decongestants

Example:

📖 Phenylephrine

Cause vasoconstriction.

Result:

☐ Increased BP

Risk Factors

These increase the likelihood of hypertension and cardiovascular disease.

✓ Age 📖 📖

✓ Diabetes Mellitus 📖 📖

✓ Dyslipidemia ☐

✓ Family History ☐

✓ Obesity 📖

✓ Physical Inactivity 📖

✓ Tobacco Use 📖

📖 Treatment Approaches

There are two major approaches.

1 📖 Non-Pharmacological Treatment

2 📖 Pharmacological Treatment 📖

☐ Non-Pharmacological Management of Hypertension

Before starting drugs, and even after drug therapy begins, lifestyle modification is essential because it lowers blood pressure, reduces cardiovascular risk, and may decrease the need for medications. 📖 ❤️

1 📖 Weight Reduction 📖

Why does weight increase BP?

Obesity causes:

◆ Increased sympathetic nervous system activity

- ◆ Increased cardiac workload
- ◆ Increased sodium retention
- ◆ Increased peripheral vascular resistance

Result:

- ↑ Blood Pressure

Benefits of Weight Loss

📅 For every 10 kg of weight lost:

- SBP decreases by approximately 5–20 mmHg

Exam Pearl 🌀

- ✓ Even a small amount of weight loss can produce significant BP reduction.

2 Regular Exercise 🏃♂️❤️

Exercise lowers BP even if the patient does not lose weight.

Benefits

- ✓ Reduces incidence of hypertension
- ✓ Protects against cardiovascular disease
- ✓ Improves endothelial function
- ✓ Decreases sympathetic activity

Recommended Exercise

🏃♂️ Aerobic exercise

Examples:

- ▼ Walking
- ▼ Running
- ▼ Cycling
- ▼ Swimming

Recommended:

🕒 30 minutes

📅 At least 3 times per week

Effect on BP

📅 Regular exercise lowers:

- SBP by approximately 4–9 mmHg

3 Dietary Changes 🍰🥗

One of the most effective lifestyle interventions.

[25/06/2026 11:20 AM] hssnd: A. Sodium Restriction □

Recommended Intake

📅 Sodium intake:

- ✓ ≤ 100 mmol/day

Equivalent to:

- ✓ 2.4 g sodium (Na)

✓ 6 g sodium chloride (NaCl)

Effect on BP

☞ Sodium restriction lowers:

□ SBP by 2–8 mmHg

B. DASH-Type Diet 🥗

Recommended Foods

🍎 Fruits

□ Vegetables

🥕 Fiber-rich foods

🥛 Low-fat dairy products

🥩 Lean proteins

Avoid

🚫 Saturated fats

🚫 Trans fats

□ Excess total fat

4 Moderation of Alcohol 🍷

Excess alcohol consumption increases BP.

Recommendation

🍷 Limit alcohol intake.

Reducing alcohol consumption can significantly lower BP in hypertensive patients.

5 Avoidance of Tobacco 🚭

Smoking is a major cardiovascular risk factor.

Long-Term Effects

📈 Arterial stiffness increases

📈 Endothelial damage occurs

📈 Risk of MI and stroke increases

📈 Cardiovascular mortality increases

High-Yield Point 🎯

☞ Each cigarette causes a transient increase in sympathetic activity and blood pressure.

□ Renin-Angiotensin-Aldosterone System (RAAS)

The Most Important Blood Pressure-Regulating System

The Renin-Angiotensin-Aldosterone System (RAAS) is a hormonal cascade that regulates:

✓ Blood pressure (BP)

✓ Blood volume

✓ Sodium balance

✓ Cardiovascular function

Why Does the Body Activate RAAS?

The body activates RAAS whenever it thinks:

- ☐ Blood pressure is too low
-  Blood volume is too low
- ☐ Kidney perfusion is reduced

The goal is simple:

Low BP



Activate RAAS



Raise BP

Step 1: Renin Release

Where is Renin Produced?

Renin is secreted by:

- ☐ Juxtaglomerular (JG) cells located in the kidney.

What Stimulates Renin Release?

Decreased Renal Blood Flow

Example:

- ☐ Hemorrhage
- ☐ Dehydration
- ☐ Hypotension

↓ Renal perfusion



JG cells stimulated



↑ Renin release

Reduced NaCl Delivery to Macula Densa

The macula densa monitors sodium delivery.

↓ NaCl reaching macula densa



Kidney thinks volume is low



↑ Renin secretion

β_1 -Receptor Stimulation

Sympathetic nervous system activation stimulates β_1 receptors on JG cells.

Sympathetic activation



β_1 stimulation



↑ Renin release

High-Yield Connection 

This is why β -blockers lower BP.

🔑 Step 2: Formation of Angiotensin I

The liver continuously produces:

☐ Angiotensinogen

Renin acts on angiotensinogen.

Angiotensinogen

+

Renin

↓

Angiotensin I

Important Point

📖 Angiotensin I is essentially inactive.

Think of it as a precursor waiting to be activated.

🔑 Step 3: Formation of Angiotensin II

Angiotensin I passes through the lungs and vascular endothelium.

There it encounters:

☐ ACE (Angiotensin-Converting Enzyme)

Also called:

☐ Kininase II

Reaction

Angiotensin I

↓ ACE

Angiotensin II

☐ Angiotensin II = The Main Villain in Hypertension

Angiotensin II is the major active molecule of RAAS.

It produces several powerful effects.

Effect ① Vasoconstriction ☐

Angiotensin II is one of the strongest vasoconstrictors in the body.

Clinical Term: Increased Afterload

What is Afterload?

Afterload = resistance against which the heart pumps.

♥ The heart must work harder.

Effect ② Aldosterone Release

Angiotensin II stimulates:

☐ Adrenal cortex (zona glomerulosa)

to release:

☐ Aldosterone

🔑 Step 4: Aldosterone Action

Aldosterone acts on:

☐ Distal tubule

☐ Collecting duct of the kidney.

What Does Aldosterone Do?

Sodium Retention

Water Retention

Potassium Excretion

[25/06/2026 11:20 AM] hssnd: Dr. Doc:

💧 ACE Inhibitors (ACEIs)

🌀 What Are ACE Inhibitors?

ACE inhibitors block:

☐ Angiotensin-Converting Enzyme (ACE)

Also called:

☐ Kininase II

Normally:

Angiotensin I

↓ ACE

Angiotensin II

ACE inhibitors stop this conversion.

☐ Remember Two Important Functions of ACE

ACE has TWO jobs:

1️⃣ Converts Angiotensin I → Angiotensin II

Ang I

↓ ACE

Ang II

2️⃣ Breaks Down Bradykinin

Bradykinin

↓ ACE

Inactive products

What Happens When ACE Is Inhibited?

ACE Inhibitor

↓

↓ Angiotensin II

+

↑ Bradykinin

These two effects explain almost ALL therapeutic effects and side effects.

I. Mechanism of Action

A. Decreased Angiotensin II Formation

Normally Angiotensin II causes:

✓ Vasoconstriction

✓ Aldosterone secretion

✓ Cardiac remodeling

After ACE inhibition:

↓ Angiotensin II

↓

Vasodilation

↓

↓ BP

Effect on Afterload

Angiotensin II normally constricts arterioles.

Ang II



Vasoconstriction



↑ Afterload

ACE inhibitors:

↓ Ang II



Vasodilation



↓ Afterload

♡ Heart pumps more easily.

Effect on Aldosterone

Normally:

Ang II



Aldosterone secretion

When ACE is blocked:

↓ Ang II



↓ Aldosterone

Effect on Preload

Less aldosterone means:

↓ Sodium retention



↓ Water retention



↓ Blood volume



↓ Preload

♡ Less work for the failing heart.

B. Increased Bradykinin

This is a VERY IMPORTANT EXAM CONCEPT 📌

Normally:

ACE



Breaks down bradykinin

After ACE inhibition:

ACE blocked



↑ Bradykinin

Bradykinin causes:



Vasodilation



Lower BP

II. Clinical Uses

1 Hypertension ☐

ACE inhibitors are first-line drugs.

Especially useful in:

- ☐ Diabetes
- ☐ Chronic kidney disease
- ☐ Heart failure

Examples:

-  Captopril
-  Enalapril
-  Lisinopril

2 Congestive Heart Failure (CHF)

One of the most important uses.

Benefits:

- ☒ Decreases preload
- ☒ Decreases afterload
- ☒ Improves cardiac output
- ☒ Reduces mortality

Why Do They Reduce Mortality?

Because they prevent:

-  Cardiac remodeling

Normally:

Ang II + Aldosterone

↓

Cardiac hypertrophy

↓

Heart failure progression

ACE inhibitors block this process.

3 Diabetic Nephropathy ☐

A favorite exam question.

ACE inhibitors:

- ☒ Reduce proteinuria
- ☒ Slow kidney damage
- ☒ Protect renal function

Why?

They reduce glomerular pressure.

Thus:

ACE inhibitor

↓

Less kidney injury

4 Post-Myocardial Infarction

After a heart attack:

ACE inhibitors help:

- ☒ Prevent remodeling
- ☒ Improve survival

- ✓ Reduce progression to heart failure

5 Scleroderma Renal Crisis

- 📌 Very High-Yield

Drug of choice:

- 💊 Captopril

III. Pharmacokinetics

Prodrugs

Many ACE inhibitors are inactive when given.

They require liver activation.

Examples:

- 💊 Enalapril
- 💊 Ramipril
- 💊 Perindopril

Active Drugs

Two important exceptions:

- ✓ Captopril
- ✓ Lisinopril

These are given in active form.

Captopril

Characteristics

- ✓ Rapid onset
- ✓ Short duration

Exam Pearl

Food decreases absorption.

Therefore:

- 🕒 Take before meals.

IV. Adverse Effects

1 Dry Cough 📌

Most common adverse effect.

Occurs in up to 30% of patients.

Why?

Because of:

ACE inhibition

↓

↑ Bradykinin

↓

Persistent dry cough

High-Yield

- 📌 Dry cough is NOT due to allergy.
- 📌 It is due to bradykinin accumulation.

2 Hyperkalemia ☐

Normally:

Aldosterone

↓
K⁺ excretion
ACE inhibitors decrease aldosterone.
Result:
↓ Aldosterone
↓
↓ K⁺ excretion
↓
Hyperkalemia
Dangerous Combination
ACE inhibitor + potassium-sparing diuretic
Example:
💊 Spironolactone
Can cause severe hyperkalemia.

3 Angioedema 🚑

Rare but potentially fatal.

What Happens?

Sudden swelling of:

👄 Lips

👅 Tongue

☐ Airway

Why?

Again due to:

↑ Bradykinin

Exam Pearl

Dry cough and angioedema are both caused by:

✅ Bradykinin accumulation

4 First-Dose Hypotension

Especially with:

💊 Captopril

Common in patients already taking diuretics.

Mechanism:

First dose

↓

Sudden vasodilation

↓

Marked BP drop

Symptoms:

🌀 Dizziness

🪤 Fainting

[25/06/2026 11:20 AM] hssnd: 5 Acute Renal Failure

This confuses many students.

Why Can ACE Inhibitors Damage Kidneys?

Normally Ang II constricts the efferent arteriole.

This helps maintain GFR.

Ang II

↓

Efferent constriction



Maintains GFR

In bilateral renal artery stenosis:

Kidneys depend on Ang II.

When ACE inhibitors are given:

ACE inhibitor



↓ Ang II



Efferent dilation



↓ GFR



Acute renal failure

V. Contraindications

⊗ Pregnancy

Absolute contraindication.

ACE inhibitors are:

✗ Teratogenic

Can cause:



Fetal renal damage



Oligohydramnios



Fetal death

⊗ Bilateral Renal Artery Stenosis

Can precipitate:

☐ Acute renal failure

⊗ Hyperkalemia

Already elevated potassium:

ACE inhibitor



Further K⁺ increase



Dangerous arrhythmias

A very high-yield pharmacology rule is:

☐ Drugs Ending in "-pril" = ACE Inhibitors



Angiotensin II Receptor Blockers (ARBs)

The “-sartan” Drugs

Examples: Losartan • Valsartan • Irbesartan • Candesartan • Telmisartan • Eprosartan



Definition

ARBs are antihypertensive drugs that selectively block the AT₁ receptor, preventing the

actions of angiotensin II.

⚙ Mechanism of Action

Normal RAAS

Renin



Angiotensinogen



Angiotensin I



Angiotensin II



AT₁ Receptor



Vasoconstriction



Aldosterone Release



↑ Blood Pressure

After ARB Administration

Angiotensin II



AT₁ Receptor **✗ BLOCKED**



No Vasoconstriction

No Aldosterone Release



↓ Blood Pressure

□ Why ARBs Are Special

Unlike ACE inhibitors, ARBs:



Block the receptor directly



Remain effective even if Ang II is formed by non-ACE pathways (e.g., chymase)



Do NOT inhibit ACE



Do NOT increase bradykinin levels



Effects of AT₁ Receptor Blockade

□ Blood Vessels

↓ Angiotensin II Action



Vasodilation



↓ Peripheral Resistance



↓ Blood Pressure



Adrenal Cortex

↓ Aldosterone



↓ Na⁺ Retention

↓ Water Retention

↓
↓ Blood Volume
↓
↓ Blood Pressure
♥ Heart
↓ Afterload
↓ Preload
↓
Improved Cardiac Function
□ Remodeling
AT₁ Blockade
↓
↓ Cardiac Hypertrophy
↓ Fibrosis
↓
Prevention of Cardiac Remodeling

Clinical Uses

1 Hypertension

- ✓ First-line antihypertensive agents
- ✓ Excellent alternative to ACE inhibitors
- ✓ Preferred when ACE inhibitors cause cough

2 Chronic Heart Failure (CHF)

Benefits:

- ♥ Decrease preload
- ♥ Decrease afterload
- ♥ Improve symptoms
- ♥ Reduce mortality
- ♥ Prevent cardiac remodeling

3 Diabetic Nephropathy

Benefits:

- Reduce proteinuria
- Slow progression of kidney disease
- Protect renal function

4 Diabetic Retinopathy

- 👁 May delay progression of retinal damage

5 Hypertension in Diabetics

- ★ Drug of choice in many diabetic hypertensive patients because of renal protection.

Adverse Effects

- Hyperkalemia
- ↓ Aldosterone
- ↓
↓ K⁺ Excretion



Hyperkalemia

Risk increases in:

- Renal impairment
- Potassium supplements
- Potassium-sparing diuretics



Postural Hypotension

More common in:

- Sodium-depleted patients
- Patients taking diuretics



Renal Dysfunction

Can precipitate acute kidney injury in:



Bilateral renal artery stenosis



Contraindications



Pregnancy

ARBs are teratogenic and may cause:



Fetal renal failure



Oligohydramnios



Fetal death



Bilateral Renal Artery Stenosis

May cause severe reduction in GFR and acute renal failure.



Hyperkalemia

Avoid in patients with already elevated potassium levels.



Losartan (Prototype ARB)

Unique Features



Most commonly used ARB



Converted in the liver to active metabolites



Long duration of action

Additional Effect



May reduce platelet aggregation by antagonizing thromboxane A₂ (TXA₂).

Diuretics



are medications that act on the kidney to modify salt and water excretion, making them essential for treating renal, cardiovascular, and endocrine disorders.



They are primarily categorized based on the specific segment of the nephron where they inhibit the reabsorption of sodium and other ions.

[25/06/2026 11:20 AM] hssnd: I. Major Classes of Diuretics



Loop Diuretics (High-Ceiling Diuretics)

Furosemide • Bumetanide • Torsemide • Ethacrynic Acid



Definition

Loop diuretics are the most powerful diuretics available.

They act on the Thick Ascending Limb (TAL) of the Loop of Henle and produce a massive excretion of sodium and water.

✧ Also called High-Ceiling Diuretics because their diuretic effect continues to increase with increasing dose.

⚙ Mechanism of Action

Site of Action

📍 Thick Ascending Limb (TAL)

Normally, this segment reabsorbs:

✓ 20–30% of filtered sodium

through the:

$\text{Na}^+/\text{K}^+/2\text{Cl}^-$ Cotransporter (NKCC2)

Tubular Lumen

↓

$\text{Na}^+/\text{K}^+/2\text{Cl}^-$ Transporter (NKCC2)

↓

$\text{Na}^+ + \text{K}^+ + \text{Cl}^-$ Reabsorption

Action of Loop Diuretics

Loop Diuretic

↓

Blocks NKCC2

↓

↓ Na^+ Reabsorption

↓ Cl^- Reabsorption

↓ Water Reabsorption

↓

Massive Diuresis

□ Effect on Calcium and Magnesium

Normally:

K^+ Recycling

↓

Positive Luminal Charge

↓

Ca^{2+} & Mg^{2+} Reabsorption

Loop diuretics abolish this positive potential.

NKCC2 Blockade

↓

Loss of Positive Lumen Potential

↓

↓ Ca^{2+} Reabsorption

↓ Mg^{2+} Reabsorption

↓

↑ Ca^{2+} & Mg^{2+} Excretion

✚ Loop diuretics increase calcium excretion.

📖 Clinical Uses

1 Acute Pulmonary Edema 🗨️❤️

First-Line Drug

Examples:

💧 Furosemide

💧 Bumetanide

Benefits:

Loop Diuretic

↓

Venodilation

↓

↓ Pulmonary Congestion

↓

Rapid Relief of Dyspnea

☆ Venodilation occurs before diuresis, providing rapid symptomatic relief.

2 Acute Heart Failure ❤️

Benefits:

✓ Reduces preload

✓ Reduces edema

✓ Improves pulmonary congestion

3 Severe Hypertension ☐

Used in:

🚑 Severe hypertension

🚑 Hypertensive emergencies

Particularly useful when:

☐ Renal impairment is present

4 Hypercalcemia ☐

Loop diuretics increase calcium excretion.

Loop Diuretic

↓

↑ Urinary Ca^{2+} Loss

↓

↓ Serum Calcium

✚ Must be combined with:

💧 Intravenous saline

to prevent dehydration and maintain effectiveness.

5 Refractory Edema 💧

Used in edema caused by:

- ♥ Heart failure
- Renal disease
- Liver cirrhosis

⚠ Adverse Effects

1 Hypokalemic Metabolic Alkalosis □

Most important adverse effect.

Why?

More sodium reaches the collecting duct.

↑ Na⁺ Delivery

↓

↑ K⁺ Secretion

↑ H⁺ Secretion

↓

Hypokalemia

+

Metabolic Alkalosis

2 Ototoxicity 🦻

🦻 Unique adverse effect of loop diuretics.

Can cause:

- 🦻 Tinnitus
- 🦻 Hearing impairment
- 🦻 Deafness (high doses)

Risk increases with:

- Rapid IV administration
- Aminoglycosides
- Renal impairment

3 Hypovolemia 💧

Because of intense diuresis:

Excess Fluid Loss

↓

Hypovolemia

↓

Hypotension

Symptoms:

- 🦻 Dizziness
- 🦻 Weakness
- 🦻 Orthostatic hypotension

4 Hyperuricemia □

Most loop diuretics:

Loop Diuretic

↓
↑ Uric Acid
↓
Hyperuricemia
↓
Gout

✧ Exception:
★ Indacrinone

5 Hypomagnesemia □
Due to increased Mg^{2+} excretion.

6 Sulfonamide Allergy 🔒
Most loop diuretics are sulfonamide derivatives.
May cause:

- Rash
- Allergic reactions

Exception
★ Ethacrynic Acid
Not a sulfonamide.
Preferred in patients with severe sulfa allergy.

💧 Thiazide Diuretics
Hydrochlorothiazide • Chlorthalidone • Indapamide • Chlorothiazide

🌀 Definition
Thiazide diuretics are moderately potent diuretics and are considered first-line drugs for hypertension.
They are sulfonamide derivatives that act mainly in the Distal Convolute Tubule (DCT).

⚙️ Mechanism of Action
📍 Site of Action
Distal Convolute Tubule (DCT)

The DCT normally reabsorbs about 5–8% of filtered sodium.

🌀 Molecular Target
 Na^+/Cl^- Cotransporter (NCC)

Normally:
Tubular Lumen
↓
 Na^+/Cl^- Cotransporter (NCC)
↓
 $Na^+ + Cl^-$ Reabsorption

Action of Thiazides
Thiazide Diuretic

↓
Blocks NCC
↓
↓ Na⁺ Reabsorption
↓ Cl⁻ Reabsorption
↓
↑ Na⁺ & Water Excretion
↓
Diuresis

□ Effect on Calcium

🏠 High-Yield Difference from Loop Diuretics

Loop Diuretics

↑ Urinary Ca²⁺ Excretion

Thiazides

↓ Urinary Ca²⁺ Excretion

[25/06/2026 11:20 AM] hssnd: Why?

Thiazides decrease intracellular sodium in DCT cells.

This enhances:

Na⁺/Ca²⁺ Exchange

↓

More Ca²⁺ Reabsorbed

↓

Less Ca²⁺ Lost in Urine

□ Key Point

Loop Diuretics

→ Lose Calcium

Thiazides

→ Save Calcium

📖 Clinical Uses

1 Hypertension □

First-Line Therapy

According to major guidelines, thiazides are among the preferred initial drugs for uncomplicated hypertension.

Benefits:

✓ Effective

✓ Inexpensive

✓ Long-term BP control

Antihypertensive Effect

↓ Na⁺

↓

↓ Blood Volume

↓

↓ Blood Pressure

Long-term:

↓ Peripheral Resistance

↓

Further BP Reduction

② Calcium Kidney Stones ☐

Recurrent Calcium Stones

Because thiazides decrease urinary calcium:

Thiazide

↓

↓ Urine Calcium

↓

↓ Calcium Stone Formation

③ Nephrogenic Diabetes Insipidus ☹

Paradoxical Effect

Although they are diuretics, thiazides reduce urine volume in nephrogenic diabetes insipidus.

Thiazide

↓

Mild Volume Depletion

↓

↑ Proximal Na^+ & Water Reabsorption

↓

Less Urine Formation

④ Mild Edema ♥

Used in:

✓ Mild heart failure

✓ Mild edema states

However:

🚫 Loop diuretics are preferred for severe edema.

⚠ Adverse Effects

① Hypokalemia ☐

Most common electrolyte abnormality.

↑ Na^+ Delivery to Collecting Duct

↓

↑ K^+ Secretion

↓

Hypokalemia

Symptoms:

⚡ Muscle weakness

⚡ Fatigue

⚡ Arrhythmias

② Hyponatremia ☐

Excess Na^+ Loss



Low Serum Sodium

Can cause:



Confusion



Dizziness



Seizures (severe cases)

3 Hypokalemic Metabolic Alkalosis

↑ K^+ Loss

+

↑ H^+ Loss

↓

Metabolic Alkalosis

4 Hyperglycemia 🧠

High-Yield

Thiazides may increase blood glucose levels.

Mechanism:

Hypokalemia

↓

Reduced Insulin Release

↓

Hyperglycemia

5 Hyperuricemia ☐

Thiazides

↓

↑ Uric Acid

↓

Gout

Patients with gout require caution.

6 Hyperlipidemia ☐

Chronic therapy may increase:



LDL



Triglycerides

7 Sulfonamide Allergy 🚫

Since most thiazides are sulfonamide derivatives:

Possible reactions:



Rash



Hypersensitivity

☐ Memory Aid

Thiazides cause:

HyperGLUC

G → Glucose ↑ 🧠

L → Lipids ↑ ☐

U → Uric acid ↑ ☐

C → Calcium retained ☐

☆ Important Drugs

💧 Hydrochlorothiazide (HCTZ)

Most commonly prescribed thiazide.

Widely used for hypertension.

💧 Chlorthalidone

Special Feature

☆ Very long acting

☆ Excellent BP control

Many studies suggest superior BP reduction compared with HCTZ.

💧 Indapamide

Unique Features

✓ Direct vasodilator effect

✓ Effective at low doses

✓ Fewer metabolic side effects

✓ Often preferred in diabetics

💧 Chlorothiazide

High-Yield

☆ Only thiazide available for IV administration.

💧 Potassium-Sparing Diuretics

Potassium-sparing diuretics are a group of diuretics that act in the cortical collecting tubule (CCT) to increase sodium excretion while decreasing potassium and hydrogen ion excretion. Unlike loop and thiazide diuretics, they do not cause hypokalemia. Instead, their major risk is hyperkalemia.

Classification

Potassium-sparing diuretics are divided into two groups:

1. Aldosterone Antagonists

◆ Spironolactone

◆ Eplerenone

2. ENaC (Sodium Channel) Blockers

◆ Amiloride

◆ Triamterene

Mechanism of Action

A. Aldosterone Antagonists

Drugs

- ◆ Spironolactone
- ◆ Eplerenone

Mechanism

These drugs enter the collecting tubule cell from the blood side (basolateral membrane) and bind to intracellular aldosterone receptors.

This prevents aldosterone from inducing the synthesis of:

- ✓ Epithelial sodium channels (ENaC)
- ✓ Na^+/K^+ ATPase pumps

Result

- ↓ Sodium reabsorption
- ↓ Water reabsorption
- ↓ Potassium secretion
- ↓ Hydrogen ion secretion

B. ENaC Blockers

Drugs

- ◆ Amiloride
- ◆ Triamterene

Mechanism

These drugs act from the luminal (urinary) side and directly block ENaC channels. Unlike spironolactone, they do not depend on aldosterone antagonism.

Result

- ↓ Sodium reabsorption
- ↓ Potassium loss
- ↓ Hydrogen ion loss

[25/06/2026 11:20 AM] hssnd: Clinical Uses

1. Prevention of Hypokalemia ☆

The most common use.

They are often combined with:

- ✓ Loop diuretics (e.g., furosemide)
- ✓ Thiazides (e.g., hydrochlorothiazide)

Purpose:

Prevent potassium depletion

Maintain normal serum potassium levels

2. Congestive Heart Failure (CHF) ♥

Important Drugs

- ◆ Spironolactone
- ◆ Eplerenone

These drugs reduce:

- ✓ Morbidity
- ✓ Mortality

Why?

Aldosterone contributes to:

Myocardial hypertrophy
Cardiac fibrosis
Ventricular remodeling
Blocking aldosterone slows disease progression.

3. Hyperaldosteronism

Primary Hyperaldosteronism

Example:

- ✓ Conn syndrome
- ✓ Secondary Hyperaldosteronism

Seen in:

- ✓ Cirrhosis
- ✓ Heart failure

Drug of Choice

🕒 Spironolactone

Adverse Effects

1. Hyperkalemia 🚫

The most serious toxicity.

Manifestations:

- ✓ Muscle weakness
- ✓ Cardiac arrhythmias
- ✓ Cardiac arrest

Risk increases when combined with:

- ◆ ACE inhibitors
- ◆ ARBs
- ◆ Potassium supplements

High-Yield Point

Never combine potassium supplements with potassium-sparing diuretics unless specifically indicated.

2. Hyperkalemic Metabolic Acidosis

Because hydrogen ion secretion is reduced:

- ✓ Potassium accumulates
- ✓ Hydrogen ions accumulate

Result:

- Hyperkalemia
- Metabolic acidosis

3. Endocrine Effects of Spironolactone

Spironolactone interacts with androgen and progesterone receptors.

In Men

- ◆ Gynecomastia
- ◆ Reduced libido
- ◆ Erectile dysfunction

In Women

- ◆ Menstrual irregularities
- ◆ Breast tenderness

Eplerenone Advantage ☆

Eplerenone is more selective for aldosterone receptors.

Therefore it causes:

Much less gynecomastia
Fewer antiandrogen effects

Pharmacokinetics

Spironolactone & Eplerenone

- ✓ Slow onset of action
- ✓ Full effect requires 24–72 hours

Amiloride & Triamterene

- ✓ Faster onset
- ✓ Duration: 12–24 hours

💧 Osmotic Diuretics

Osmotic diuretics are drugs that increase urine output by holding water inside the renal tubules through osmotic forces 💧.

Unlike loop or thiazide diuretics, they do not block ion transporters. Instead, they act as non-reabsorbable solutes that prevent water reabsorption.

✳ Examples

📌 Mannitol (prototype)

💧 Glycerol

☐ Urea

💧 Isosorbide

⚙ Mechanism of Action

Step 1: Filtration

Osmotic diuretics are:

- ✓ Freely filtered by the glomerulus
- ✗ Poorly reabsorbed by renal tubules

Step 2: Water Retention in Tubules

Since the drug remains inside the tubular lumen:

💧 Water follows the osmotic gradient and stays in the tubule.

Result

- ↑ Urine volume
- ↑ Water excretion
- ↓ Water reabsorption

📍 Site of Action

1 Proximal Convulated Tubule (PCT) ☆

This is the major site of action.

Normally:

☐ Large amounts of sodium and water are reabsorbed here.

Mannitol prevents water from following sodium.

2 Descending Limb of Loop of Henle

This segment is highly permeable to water Δ .

Mannitol reduces water reabsorption here.

3 Collecting Tubule

Additional inhibition of water reabsorption occurs.

✱ Mannitol (Prototype Drug)

Characteristics

- ☒ Most commonly used osmotic diuretic
- ☒ Pharmacologically inert
- ☒ Poorly reabsorbed
- ☒ Remains in tubular fluid

Route of Administration

 IV infusion only

☒ Not given orally for diuretic purposes

Mannitol = Intravenous Drug

Clinical Uses

1 Increased Intracranial Pressure (ICP) ☐

Drug of Choice ☆

Used in:

 Head injury

☐ Cerebral edema

 Neurosurgical emergencies

Mechanism

☒ Mannitol increases plasma osmolarity.

Water moves:

☐  ☐

(Brain tissue → Bloodstream)

Result

☐  Brain swelling

☐  Intracranial pressure

2 Acute Glaucoma

Mannitol draws water from the eye into the bloodstream.

  ☐

Result

↓ Intraocular pressure (IOP)

Used in:

👁️ Acute angle-closure glaucoma

👤 ☐ Before eye surgery

3️⃣ Prevention of Acute Renal Failure ☐

May be used to maintain urine flow in:

🚑 Trauma

☐ Shock

🏥 Major surgery

Benefit

↑ Tubular flow

↓ Tubular obstruction

4️⃣ Promotion of Toxin Elimination ☠️

Sometimes used to increase urinary excretion of certain toxins.

[25/06/2026 11:20 AM] hssnd: ⚠️ Adverse Effects

1️⃣ Expansion of Extracellular Fluid Volume 📈

Before diuresis occurs:

Mannitol pulls water from cells into blood vessels.

☐ 🔄 ☐ → ☐

Result

↑ Plasma volume

↑ Cardiac workload ❤️

2️⃣ Pulmonary Edema 📈

Particularly dangerous in patients with heart disease.

Why?

↑ Intravascular volume

↓ Cardiac reserve

→ Fluid accumulates in lungs

Symptoms

😓 ⇐ Shortness of breath

😓 Respiratory distress

👤 Pulmonary edema

3️⃣ Dehydration 💧

Excessive diuresis may cause:

☐ Thirst

🌀 Dizziness

📉 Hypotension

4️⃣ Electrolyte Disturbances ☐

May produce:

☐ ↓ Hyponatremia (early)

☐ ↑ Hypernatremia (later)

depending on fluid shifts.

⊘ Contraindications

♥ Congestive Heart Failure (CHF)

Why?

☐ ↑ Plasma volume may worsen heart failure.

🫁 Pulmonary Edema

May aggravate lung congestion.

💧 Severe Dehydration

Can worsen volume depletion.

☐ Active Intracranial Hemorrhage

Generally avoided except in specific neurosurgical situations.

💊 Calcium Channel Blockers (CCBs)

Calcium Channel Blockers (CCBs) are drugs that block L-type voltage-gated calcium channels in:

♥ Cardiac muscle

☐ Vascular smooth muscle

By reducing calcium entry into cells, they produce:

☒ Vasodilation

☒ Reduced cardiac workload

☒ Decreased oxygen demand

☒ Slower cardiac conduction

⚙️ Mechanism of Action

Normally:

Ca^{2+} enters through L-type channels

↓

Smooth muscle contraction

↓

Vasoconstriction

CCBs block calcium entry:

CCBs

↓

↓ Ca^{2+} influx

↓

Smooth muscle relaxation

↓

Vasodilation

↓

↓ Blood Pressure

Classification of CCBs

CCBs are divided into 3 major groups:

1 Dihydropyridines (DHPs)

Examples

-  Nifedipine
-  Amlodipine
-  Felodipine
-  Nicardipine
-  Nimodipine

Main Action

□ Blood vessels >>> Heart

They are potent vasodilators.

Remember

👉 "DHP = Dilate vessels"

2 Phenylalkylamines

Example

 Verapamil ☆

Main Action

♥ Heart >>> Blood vessels

Effects:

- ↓ Heart rate (negative chronotropic)
- ↓ Contractility (negative inotropic)
- ↓ AV conduction (negative dromotropic)

3 Benzothiazepines

Example

 Diltiazem

Main Action

♥ + □

- ✓ Acts on both heart and blood vessels.
- ✓ Its cardiac depression is less than verapamil.

Clinical Uses

1 Hypertension □

First-Line Antihypertensives

CCBs are recommended by major guidelines.

Especially effective in:

-  Elderly patients
-  African ancestry patients

Why?

CCB



Vasodilation



↓ Peripheral Resistance



↓ Blood Pressure

2 Angina Pectoris ♥

A. Stable (Effort) Angina

CCBs decrease:

♥ Heart rate

♥ Contractility

♥ Oxygen demand

Result:

☑ Less anginal pain

B. Vasospastic (Prinzmetal) Angina

CCBs relax coronary arteries.

Coronary Vasospasm



CCB



Coronary Vasodilation



↑ Oxygen Supply

Drug of Choice ☆

💊 Calcium Channel Blockers

3 Supraventricular Tachycardia (SVT) ⚡

Important Drugs

💊 Verapamil

💊 Diltiazem

Mechanism

They slow conduction through the AV node.

Result:

♥ Slower ventricular rate

♥ Termination of SVT

4 Migraine Prophylaxis 🤢

Certain CCBs help prevent migraine attacks.

5 Raynaud Phenomenon 🌸

Raynaud disease causes excessive vasoconstriction of fingers.

CCBs:

☐ Dilate peripheral vessels

👉 Improve blood flow

6 Subarachnoid Hemorrhage ☐

Drug

🔑 Nimodipine ☆

Use

Prevents cerebral vasospasm after:

☐ Subarachnoid hemorrhage

High-Yield Exam Point

👉 Nimodipine = Brain Vessel Protector

⚠ Adverse Effects

1 Constipation 📄

Most common with:

🔑 Verapamil

2 Ankle (Pretibial) Edema ☐

Due to arteriolar dilation.

Symptoms:

☐ Swollen ankles

☐ Leg edema

3 Flushing 🟡

Caused by vasodilation.

4 Headache 😖

Due to cerebral vasodilation.

5 Dizziness 🌀

Due to reduced blood pressure.

6 Bradycardia ❤️ ☑

Most common with:

🔑 Verapamil

🔑 Diltiazem

7 AV Block ⚡

Particularly with:

🔑 Verapamil

Risk increases when combined with:

🔑 Beta blockers

8 Heart Failure ❤️ 🏠

Verapamil decreases cardiac contractility.

Can worsen existing heart failure.

[25/06/2026 11:20 AM] hssnd: ⚠ Reflex Tachycardia

Mainly with Short-Acting Nifedipine

Mechanism:

Rapid Vasodilation



Sudden BP Fall



Baroreceptor Activation



Reflex Tachycardia

Consequences



Increased myocardial oxygen demand



Worsening angina



Increased MI risk



Contraindications



Heart Failure 

Especially:



Verapamil



Diltiazem

May worsen cardiac function.



Sick Sinus Syndrome 

Risk of severe bradycardia.



Advanced AV Block 

Can worsen conduction abnormalities.



Combination with Beta Blockers 

Example:



Verapamil + Propranolol

May cause:



Severe bradycardia



Complete heart block



Cardiogenic shock



α_1 -Adrenergic Antagonists (Selective α_1 -Blockers)

Selective α_1 -adrenergic antagonists (α_1 -blockers) are sympatholytic drugs that selectively block α_1 receptors on vascular smooth muscle and other tissues.

They are mainly used for:



Hypertension



Benign Prostatic Hyperplasia (BPH)

Unlike nonselective α -blockers, they produce less reflex tachycardia and are better tolerated for long-term use.



Drugs

Prototype Drug 



Prazosin

Duration of action:

 6–8 hours

Long-Acting Agents

 Terazosin

 Doxazosin

Advantages:

- ☒ Longer duration
- ☒ Once-daily dosing
- ☒ Better patient compliance

Clinical Uses

1 Hypertension ☐

Used in:

- ☒ Mild hypertension
- ☒ Moderate hypertension

Particularly useful when patients have:

- ☒  BPH
- ☐ Diabetes
- ☐ Dyslipidemia
- ☐ Gout

Why They Lower BP

They cause:

- ☐ Arteriolar dilation
- ☐ Venous dilation

Result:

- ☐ ↓ Peripheral vascular resistance
- ☐ ↓ Blood pressure

2 Benign Prostatic Hyperplasia (BPH)  ☆

α_1 receptors are abundant in:

- ☒  Prostate
- ☒  Bladder neck

Blockade causes:

- ☒ Smooth muscle relaxation
- ☒ Reduced urinary obstruction
- ☒ Improved urinary stream

Symptoms Improved

-  Urinary frequency
-  Nocturia
-  Hesitancy
-  Weak stream

Preferred Drugs

- 🕒 Prazosin
- 🕒 Terazosin
- 🕒 Doxazosin

(Clinically, tamsulosin is often preferred because it is more selective for α_1A receptors in the prostate.)

3 Hypertension During Pregnancy 🧑

Prazosin is considered relatively safe in pregnancy.

May be used when antihypertensive treatment is required.

✳ Metabolic Advantages

One major advantage of α_1 -blockers is that they do not worsen metabolic disorders.

In Diabetes 🧑

Unlike some other antihypertensives:

- ✓ No significant effect on blood glucose
- ✓ Safe in diabetic patients

In Gout ☐

They do not increase uric acid levels.

- ✓ Safe in gout patients

Effects on Lipid Profile ☐

α_1 -blockers may improve lipid metabolism.

Effects

- ↓ LDL ("bad cholesterol")
- ↓ Triglycerides
- ↑ HDL ("good cholesterol")

⚠ Adverse Effects

1 First-Dose Orthostatic Hypotension 🧑 ☆

Most Important Adverse Effect

After the first dose:

- ☐ Standing up suddenly
- ↓ Marked drop in BP
- 🌀 Dizziness
- 🌀 Fainting (syncope)

Prevention

- 💡 Start with a low dose.
- 💡 Give first dose at bedtime.

2 Postural Hypotension

Due to excessive vasodilation.

Symptoms:

-  Dizziness
-  Lightheadedness
- ☐ Weakness

3 Reflex Tachycardia

Less common than with nonselective α -blockers, but may occur.

Mechanism:

↓ Blood Pressure

↓

Baroreceptor Reflex

↓

↑ Heart Rate

4 Nasal Congestion

Vasodilation of nasal blood vessels may cause:

-  Nasal stuffiness

5 Headache

Due to vasodilation.

Contraindications / Cautions

Use cautiously in:

-  Elderly patients
- ☐ Patients prone to orthostatic hypotension
-  Individuals at risk of falls

β -Adrenergic Blockers (β -Blockers)

β -Blockers are drugs that block β -adrenergic receptors, thereby reducing the effects of the sympathetic nervous system on the:

-  Heart
- ☐ Kidneys
- ☐ Central nervous system

They are widely used in:

- ☐ Hypertension
-  Angina
-  Heart Failure
-  Arrhythmias

Mechanism of Action

β -blockers lower blood pressure and cardiac workload through multiple mechanisms.

[25/06/2026 11:20 AM] hssnd:  Decrease Cardiac Output 

Normal β_1 Stimulation

β_1 Activation

↓

↑ Heart Rate

↑ Contractility

↓

↑ Cardiac Output
After β_1 Blockade

β_1 Blockade

↓

↓ Heart Rate

↓ Contractility

↓

↓ Cardiac Output

↓

↓ Blood Pressure

Effects

♡ Negative chronotropic effect
(↓ heart rate)

♡ Negative inotropic effect
(↓ contractility)

2 Decrease Renin Release □

β_1 receptors are present on:

□ Juxtaglomerular (JG) cells

Normally:

β_1 Stimulation

↓

↑ Renin Release

↓

↑ Angiotensin II

↓

↑ Aldosterone

↓

↑ Blood Pressure

β -blockers inhibit this pathway:

β_1 Blockade

↓

↓ Renin

↓ Angiotensin II

↓ Aldosterone

↓

↓ Blood Pressure

3 Reduce Sympathetic Activity □

Some β -blockers reduce central sympathetic outflow.

Result:

✓ Lower blood pressure

✓ Reduced cardiac stress

📖 Classification of β -Blockers

1 Nonselective β -Blockers

Block:

☒ β_1 receptors

☒ β_2 receptors

Examples

 Propranolol ☆

 Timolol

Effects

♥ Heart effects

□ Bronchoconstriction

⚠ Greater risk in asthmatic patients

2 β_1 -Selective (Cardioselective) Blockers

Primarily block β_1 receptors.

Examples

 Atenolol

 Metoprolol

 Esmolol

 Bisoprolol

Advantages

☒ Less bronchospasm

☒ Safer in asthma

☒ Safer in diabetes

Memory Tip □

Most β_1 -selective drugs end in:
"-olol"

Examples:

 Atenolol

 Metoprolol

 Bisoprolol

3 Combined $\alpha + \beta$ Blockers

Block:

☒ α_1 receptors

☒ β receptors

Examples

 Carvedilol

 Labetalol

Additional Effect

α_1 Blockade

↓

Vasodilation



Further BP Reduction

4 Partial Agonists (ISA)

ISA = Intrinsic Sympathomimetic Activity

Examples



Pindolol



Acebutolol

Advantage

Less likely to cause:



Severe bradycardia



Excessive cardiac depression



Clinical Uses

1 Hypertension ☐

Used for:



Hypertension with angina



Hypertension after MI

Not preferred as first-line therapy for uncomplicated hypertension.



Angina Pectoris

Stable (Effort) Angina

Mechanism

β -blockers reduce:



Heart rate



Contractility



Blood pressure

Result:



Myocardial oxygen demand



Anginal attacks

Double Product Concept

Heart Rate $\times$ Systolic BP

This reflects myocardial oxygen consumption.

β -blockers decrease both components.



Variant (Prinzmetal) Angina

β -blockers are contraindicated.

Why?

β_2 Blockade



Unopposed α Activity



Coronary Vasospasm



Worsened Angina

3 Chronic Heart Failure

Important Drugs

-  Carvedilol
-  Metoprolol
-  Bisoprolol

Benefits

- ☒ Reduce mortality
- ☒ Slow disease progression
- ☒ Reverse cardiac remodeling

Start with:

-  Very low doses

Then increase gradually.

-  Contraindicated In

Acute decompensated heart failure.

4 Arrhythmias

Class II Antiarrhythmics

Useful for:

-  Supraventricular tachycardia (SVT)
-  Atrial fibrillation
-  Atrial flutter

Mechanism

β_1 Blockade

↓

↓ AV Node Conduction

↓

↓ Heart Rate

Special Drug

-  Esmolol

Characteristics:

- ☒ Ultra-short acting
- ☒ IV administration

Used for acute arrhythmias.

Adverse Effects

1 Bradycardia

Excessive slowing of heart rate.

Symptoms:

-  Dizziness
-  Fatigue
-  Syncope

2 Heart Block ⚡

β -blockers slow AV node conduction.

May cause:

- ✗ AV block
- ✗ Severe bradycardia

3 Bronchospasm ☐ ⚠

Mainly with:

💊 Propranolol

💊 Timolol

Mechanism

β_2 Blockade

↓

Bronchoconstriction

Dangerous in:

- ☐ Asthma
- ☐ COPD

4 Hyperglycemia 🧠

Can impair glucose regulation.

5 Masking of Hypoglycemia ⚠

Normally hypoglycemia causes:

♥ Tachycardia

😓 Tremor

⚡ Palpitations

β -blockers hide these warning signs.

Important in:

🧠 ☐ Diabetic patients on insulin

6 Dyslipidemia ☐

May cause:

↑ Triglycerides

↓ HDL cholesterol

7 CNS Effects ☐

Especially with lipophilic drugs (e.g., propranolol).

Symptoms:

😴 Sedation

🌙 Nightmares

🌀 Sleep disturbances

8 Sexual Dysfunction ☐ ⚠

May cause:

- ✗ Erectile dysfunction
- ✗ Reduced libido

Withdrawal Syndrome

Never Stop Suddenly

Chronic β -blocker therapy causes receptor upregulation.

Abrupt withdrawal may produce:

♡ Rebound tachycardia

□ Rebound hypertension

♡ Angina

♡ Myocardial infarction

[25/06/2026 11:20 AM] hssnd: Rule

☑ Taper gradually

✗ Do not stop abruptly

⊘ Contraindications

1  Asthma & COPD □

Especially:

🔇 Propranolol

🔇 Timolol

2  Variant (Prinzmetal) Angina ♡

May worsen coronary vasospasm.

3  Acute Decompensated Heart Failure ♡

Can further reduce cardiac output.

4  Severe Bradycardia ♡ 

Risk of cardiac arrest.

5  Advanced AV Block ⚡

May worsen conduction abnormalities.

Direct Vasodilators

Vasodilators are drugs that lower blood pressure by directly relaxing vascular smooth muscle, independent of the autonomic nervous system.

Main Effects

□ ↓ Blood Pressure

♡ ↓ Cardiac Workload

□ ↓ Myocardial Oxygen Demand

Classification of Vasodilators

1  Arteriolar Dilators (↓ Afterload)

These drugs mainly dilate arterioles (resistance vessels).

Examples

- 💊 Hydralazine
- 💊 Minoxidil
- 💊 Diazoxide

Effect

Arteriolar Dilation



↓ Peripheral Resistance



↓ Afterload



↓ Blood Pressure

2 Venodilators (↓ Preload)

These drugs mainly dilate veins (capacitance vessels).

Example

- 💊 Nitroglycerin

Effect

Venous Dilation



Venous Pooling



↓ Venous Return



↓ Preload

3 Mixed Arterial + Venous Dilators

These drugs dilate both arteries and veins.

Example

- 💊 Sodium Nitroprusside

Effect



↓ Preload



↓ Afterload



Rapid BP reduction



Hydralazine

Hydralazine is an orally active direct arteriolar vasodilator.

Key Characteristic



Dilates arterioles much more than veins.

Therefore:



Markedly reduces afterload



Minimal effect on preload



Mechanism of Action

Hydralazine relaxes vascular smooth muscle through two important mechanisms.



Nitric Oxide (NO) Release

Hydralazine stimulates endothelial cells to release:

☐ Nitric Oxide (NO)

Pathway

Hydralazine



↑ Nitric Oxide (NO)



↑ Guanylyl Cyclase



↑ cGMP



Smooth Muscle Relaxation



Vasodilation

Important Point



Requires intact endothelium

② Opening of Potassium Channels

Hydralazine opens K^+ channels.

Result

K^+ Efflux



Hyperpolarization



Less Smooth Muscle Contraction



Vasodilation



Clinical Uses



Hypertension ☐

Used for:



Moderate hypertension



Severe hypertension

Usually NOT given alone.

② Chronic Heart Failure 

Special Combination



Hydralazine



Isosorbide Dinitrate

Benefits



Reduces mortality



Improves symptoms



Particularly effective in African American patients

Why?

Hydralazine:



↓ Afterload

Isosorbide dinitrate:

□ ↓ Preload

Together:

♥ Reduce cardiac workload significantly

3] Hypertensive Emergency in Pregnancy 🧑🏻 ⭐

Hydralazine can be given:

📌 IV

for rapid blood pressure control.

Traditionally considered a drug of choice for:

🧑🏻 Pregnancy-induced hypertension

🧑🏻 Preeclampsia

⚠️ Compensatory Responses

Hydralazine causes marked vasodilation.

The body tries to compensate.

1] Reflex Tachycardia ♥ 1]

Mechanism

Hydralazine

↓

↓ Blood Pressure

↓

Baroreceptor Activation

↓

↑ Sympathetic Activity

↓

↑ Heart Rate

Result

♥ Tachycardia

♥ Palpitations

♥ Increased contractility

2] Salt and Water Retention 💧

Mechanism

↓ Blood Pressure

↓

↑ Renin Release

↓

↑ RAAS Activation

↓

↑ Aldosterone

↓

Salt & Water Retention

Consequence

💧 Edema

🧑🏻 Weight gain

Why Combination Therapy Is Necessary

Hydralazine is rarely used alone.

Add a β -Blocker

 Metoprolol

 Atenolol

Purpose:

☒ Prevent reflex tachycardia

Add a Diuretic

 Furosemide

 Hydrochlorothiazide

Purpose:

☒ Prevent fluid retention

Classic Combination

Hydralazine

+

β -Blocker

+

Diuretic

Adverse Effects

1 Headache 

Most common.

Due to cerebral vasodilation.

2 Flushing 

Caused by cutaneous vasodilation.

3 Tachycardia 

Very common.

Due to baroreceptor reflex activation.

4 Lupus-Like Syndrome  

High-Yield Adverse Effect

Occurs with:

 High doses

 Long-term therapy

Usually >200 mg/day

Symptoms

 Fever

☐ Joint pain

 Fatigue

🦋 Positive ANA

Important Feature

☑ Reversible after drug withdrawal

☐ Hydralazine Metabolism

Hydralazine is metabolized by:

☐ Acetylation in the liver

[25/06/2026 11:20 AM] hssnd: Genetic Variation

Two groups exist:

🐢 Slow acetylators

⚡ Fast acetylators

Slow Acetylators

Higher drug levels

⬆ Greater efficacy

⬆ Greater toxicity

⬆ Higher risk of lupus syndrome

👶 Hydralazine in Pregnancy

Hydralazine is considered:

☑ Safe in pregnancy

Used for:

👶 Severe hypertension

👶 Preeclampsia

👶 Eclampsia-related hypertension

💧 Minoxidil

Minoxidil is a very potent, orally active direct arteriolar vasodilator used primarily for the treatment of severe or resistant hypertension.

Key Characteristic ☆

☑ Dilates arterioles only

✗ Little or no effect on veins

Therefore:

☐ ↓ Peripheral vascular resistance

☐ ↓ Blood pressure

♥ ↓ Afterload

⚙ Mechanism of Action

1️⃣ Prodrug Activation

Minoxidil itself is inactive.

In the liver:

Minoxidil

↓ (Liver)

Minoxidil Sulfate

↓

Active Form

☆ Minoxidil sulfate is the active metabolite.

2 Potassium Channel Opening

The active metabolite opens ATP-sensitive K^+ channels in vascular smooth muscle.

What Happens?

K^+ Channels Open

↓

K^+ Leaves Cell

↓

Hyperpolarization

↓

↓ Calcium Entry

↓

Smooth Muscle Relaxation

↓

Arteriolar Vasodilation

Final Effect

- ☐ Massive reduction in peripheral resistance
- ☐ Significant fall in blood pressure

Clinical Uses

1 Severe Hypertension ☆

Main Use

Used when hypertension is:

- ✗ Resistant to standard drugs
- ✗ Not controlled by multiple antihypertensives

Why?

Because Minoxidil is one of the most powerful oral vasodilators available.

2 Hypertension with Chronic Kidney Disease ☐

Special Advantage

Minoxidil remains effective even in:

- ☐ Chronic renal failure
- ☐ Advanced kidney disease

Because:

- ✓ Plasma levels are not significantly affected by renal impairment.

3 Alopecia (Hair Loss) ☆

Topical Minoxidil is widely used for:

-  Male-pattern baldness
-  Female-pattern hair loss

Why Does Hair Grow?

Minoxidil:

- Increases blood flow to hair follicles
- Prolongs the anagen (growth) phase
- 🌀 Stimulates hair growth

⚠️ Compensatory Responses

Because Minoxidil causes powerful vasodilation, the body responds aggressively.

1 Reflex Tachycardia ❤️ ↑

Mechanism

Minoxidil

↓

↓ Blood Pressure

↓

Baroreceptor Reflex

↓

↑ Sympathetic Activity

↓

↑ Heart Rate

Consequences

❤️ Tachycardia

❤️ Palpitations

❤️ Increased myocardial oxygen demand

2 Salt and Water Retention 💧

Mechanism

↓ Blood Pressure

↓

↑ Renin Release

↓

↑ RAAS Activation

↓

↑ Aldosterone

↓

Salt & Water Retention

Consequences

💧 Edema

🏋️ Weight gain

💊 Combination Therapy

Minoxidil should almost never be used alone.

Add a β -Blocker ❤️

Examples:

💊 Atenolol

💊 Metoprolol

Purpose:

✓ Prevent reflex tachycardia

Add a Diuretic 💧

Examples:

💧 Furosemide

💧 Hydrochlorothiazide

Purpose:

✓ Prevent fluid retention

Classic Regimen

Minoxidil

+

β-Blocker

+

Diuretic

⚠️ Adverse Effects

1️⃣ Hypertrichosis (Excess Hair Growth) 🧑🏻 ⭐

Most Characteristic Adverse Effect

Excessive hair growth may occur on:

🧑🏻 Face

🧑🏻 Arms

🧑🏻 Back

🧑🏻 Face (especially troublesome in women)

Clinical Significance

This adverse effect led to the development of:

❑ Topical Minoxidil for baldness treatment

2️⃣ Reflex Tachycardia ❤️

Very common.

May worsen:

❤️ Angina

❤️ Ischemic heart disease

3️⃣ Fluid Retention 💧

Can cause:

❑ Peripheral edema

⚖️ Weight gain

4️⃣ Pericardial Effusion ❤️ ⚠️

Rare but serious.

May lead to:

❑ Cardiac tamponade

Especially in patients with:

❑ Renal disease

5 Headache 🤔

Due to vasodilation.

⊘ Contraindications / Cautions

Use cautiously in:

💔 Coronary artery disease

💔 Angina

💔 Heart failure

Because reflex tachycardia may increase cardiac workload.

💧 Diazoxide

Diazoxide is a potent direct arteriolar vasodilator used mainly for hypertensive emergencies.

Although chemically related to thiazide diuretics, it has:

✗ No diuretic activity

⚙ Mechanism of Action

Potassium (K^+) Channel Opener ☆

Diazoxide opens ATP-sensitive potassium channels in:

☐ Vascular smooth muscle

☐ Pancreatic β -cells

[25/06/2026 11:20 AM] hssnd: Effect on Blood Vessels

Diazoxide

↓

Opens K^+ Channels

↓

K^+ Efflux

↓

Hyperpolarization

↓

↓ Calcium Entry

↓

Smooth Muscle Relaxation

↓

Arteriolar Vasodilation

↓

↓ Blood Pressure

Result

✓ Powerful arteriolar dilation

✓ Reduced peripheral vascular resistance

✓ Reduced blood pressure

☐ Effect on the Pancreas

Diazoxide also acts on pancreatic β -cells.

Normal Situation

Glucose

↓
Insulin Release
↓
↓ Blood Sugar
With Diazoxide
Diazoxide
↓
K⁺ Channel Opening
↓
↓ Insulin Release
↓
↑ Blood Sugar
Result
☞ Hyperglycemia

Clinical Uses

1 Hypertensive Emergencies ☆

Main Use

Given intravenously for:

☐ Malignant hypertension

☞ Eclampsia

☐ Severe hypertensive crisis

Why Useful?

Because it:

⚡ Acts rapidly

⚡ Produces marked BP reduction

⚡ Effective in 75–85% of patients

2 Insulinoma (Hypoglycemia) ☞

Insulinoma

A tumor that secretes excessive insulin.

Too Much Insulin

↓

Severe Hypoglycemia

Diazoxide Helps By

Diazoxide

↓

↓ Insulin Release

↓

↑ Blood Glucose

Therefore:

☞ Oral diazoxide can be used for insulinoma-associated hypoglycemia.

Adverse Effects

1 Hyperglycemia ☞ ☆

Most Characteristic Adverse Effect

Because:

Diazoxide

↓

↓ Insulin Secretion

↓

↑ Blood Glucose

□ Diazoxide causes hyperglycemia
while

□ Insulin lowers glucose

2 Salt and Water Retention 💧

Mechanism

↓ Blood Pressure

↓

RAAS Activation

↓

↑ Aldosterone

↓

Na⁺ & Water Retention

Result

□ Edema

⚖️ Weight gain

3 Reflex Tachycardia ❤️

Mechanism

Vasodilation

↓

↓ Blood Pressure

↓

Baroreceptor Reflex

↓

↑ Sympathetic Activity

↓

↑ Heart Rate

Result

❤️ Tachycardia

❤️ Palpitations

4 Myocardial Ischemia ❤️

Reflex tachycardia increases:

❤️ Heart rate

❤️ Oxygen demand

This may precipitate:

❤️ Angina

❤️ Myocardial ischemia

Especially in patients with coronary artery disease.

5 Hyperuricemia ☐

Can increase:

☐ Serum uric acid

May worsen:

☐ Gout

6 Excessive Hypotension ☐ ☐ ↓

Particularly with IV administration.

May cause:

🌀 Dizziness

🌀 Syncope

⚠️ Severe hypotension

👶 Effect on Uterus

Diazoxide relaxes uterine smooth muscle.

Diazoxide

↓

Uterine Relaxation

↓

May Delay Labor

Clinical Importance

⚠️ May stop or delay labor.



Route of Administration

Hypertensive Emergencies



Intravenous (IV)

Rapid onset.

Insulinoma



Oral

Longer-term treatment.



Duration of Action

After IV Administration



Lasts several hours

Longer acting than many emergency antihypertensive drugs.



Sodium Nitroprusside (SNP)

Sodium Nitroprusside is an extremely potent, rapid-acting arterial and venous vasodilator used exclusively by intravenous (IV) infusion for the treatment of hypertensive emergencies.

Key Characteristic ☆

Unlike most vasodilators:



Dilates arteries (arterioles)



Dilates veins (venules)

Therefore:

- ☐ ↓ Afterload
- ☐ ↓ Preload
- ☐ ↓ Blood Pressure

☐ Classification

Mixed Vasodilator

Sodium Nitroprusside

↓

Arteriolar Dilation

+

Venous Dilation

Result

- ☐ ↓ Peripheral resistance (↓ Afterload)
- ☐ ↓ Venous return (↓ Preload)
- ♥ Improved cardiac performance

⚙ Mechanism of Action

Step 1: Release of Nitric Oxide (NO)

Sodium nitroprusside spontaneously releases:

- ☐ Nitric Oxide (NO)

Step 2: Activation of Guanylyl Cyclase

NO

↓

Guanylyl Cyclase Activation

↓

↑ cGMP

Step 3: Smooth Muscle Relaxation

↑ cGMP

↓

Myosin Light Chain Dephosphorylation

↓

Smooth Muscle Relaxation

↓

Vasodilation

Final Result

Sodium Nitroprusside

↓

↑ NO

↓

↑ cGMP

↓

Arterial + Venous Dilation

↓

↓↓ Blood Pressure

☐ Nitroprusside dilates BOTH arteries and veins.

□ Nitrates mainly dilate veins.

Clinical Uses

1 Hypertensive Emergencies ☆

Drug of Choice

Used for:

- Malignant hypertension
- Hypertensive crisis
- Severe hypertensive emergencies

Why?

IV Nitroprusside

↓

Immediate Vasodilation

↓

Rapid BP Reduction

[25/06/2026 11:20 AM] hssnd:  2 Acute Heart Failure ♥ ☆

Used in:

- ♥ Acute decompensated heart failure
- 💧 Pulmonary edema
- ♥ Severe ventricular dysfunction

Benefits

Venous Dilation

Venodilation

↓

↓ Venous Return

↓

↓ Preload

Arterial Dilation

Arteriolar Dilation

↓

↓ Afterload

↓

Easier Ventricular Ejection

Overall Effect

- ♥ Improved cardiac output
- ♥ Reduced congestion
- ♥ Improved ventricular performance

3 Aortic Dissection ♥

Often combined with:

💧 Esmolol

or

💧 Labetalol

Why Combination?

Nitroprusside lowers BP rapidly.

β -blocker prevents:

- ♥ Reflex tachycardia
- ♥ Increased shear stress

Aortic Dissection



Esmolol + Nitroprusside



Rapid BP Control

 Route of Administration
Only Route

 Intravenous Infusion

Why IV Only?

Because:

- ⚡ Very rapid onset
- ⚡ Very short duration
- ⚡ Easily titrated

⚠ Adverse Effects

1 Excessive Hypotension ☐ ↓

Most Common Acute Problem

Because it is extremely potent.

May cause:

- 🌀 Dizziness
- 🌀 Syncope
- ⚠ Circulatory collapse

2 Headache 🤔

Due to cerebral vasodilation.

3 Nausea & Vomiting 🤢

Common during infusion.

4 Reflex Tachycardia ♥

Can occur due to sudden BP reduction.

💀 Cyanide Toxicity ☆☆☆

Most Important Toxicity

Nitroprusside molecule contains:

💀 Cyanide (CN^-)

Mechanism

Nitroprusside



Releases Cyanide



Accumulation



Cellular Hypoxia

Symptoms



Confusion



Weakness

☐ Metabolic acidosis



Arrhythmias



Death (severe cases)

Risk Factors



Prolonged infusion



High doses

☐ Renal impairment

☐ Hepatic impairment



Thiocyanate Toxicity ☆

Normally:

Cyanide



(Liver)

Thiocyanate



Kidney Excretion

If accumulation occurs:



Thiocyanate toxicity develops.

Symptoms



Psychosis



Confusion

☐ Nausea



Muscle spasms



Hypothyroidism

Why?

Thiocyanate is:



Antithyroid

It inhibits iodine utilization.

Result

↑ Thiocyanate



↓ Thyroid Hormone Production



Hypothyroidism



Salt and Water Retention

Like other vasodilators:

↓ Blood Pressure



RAAS Activation



Salt & Water Retention

Therefore Often Combined With



Diuretics



β-blockers



Contraindications



Pregnancy



Absolute Contraindication

Reason:



Risk of fetal cyanide toxicity

Use Cautiously In



Renal failure



Liver disease

Because toxicity accumulates more easily.



Ganglionic Blocking Agents (Ganglion Blockers)

Ganglionic blockers are sympathoplegic drugs that block transmission in all autonomic ganglia (both sympathetic and parasympathetic).



Historically used for hypertension, but now largely obsolete because of severe side effects.



Mechanism of Action

Target



Nicotinic neuronal (N_n) receptors in autonomic ganglia

These receptors are found in:



Sympathetic ganglia



Parasympathetic ganglia

Normal Transmission

Preganglionic neuron



ACh



N_n receptor



Postganglionic neuron



Target organ

After Ganglion Blockade

Preganglionic neuron



ACh



N_n blocked



No transmission



Autonomic activity lost

□ Why Blood Pressure Falls?

The sympathetic nervous system normally maintains vascular tone.

Ganglion Blocker



↓ Sympathetic Activity



Vasodilation



↓ Peripheral Resistance



↓ Blood Pressure

✓ Powerful antihypertensive effect

⚠ Major Problem

These drugs also block:

✗ Parasympathetic ganglia

Therefore:

✗ Entire autonomic nervous system is affected

This explains their many toxicities.



Important Drugs

1 Hexamethonium

Prototype Drug ☆

Characteristics:

✓ Blocks N_n receptors

✓ No CNS penetration

✓ Oral and parenteral use

✗ Obsolete today

2 Trimethaphan

Short-Acting Ganglion Blocker

Characteristics:

⚡ Rapid onset

⚡ Very short duration

💉 Given IV

Clinical Use

🏠 Hypertensive emergencies (historically)

🏠 Controlled hypotension during surgery

🏠 Occasionally used with sodium nitroprusside in aortic dissection

3 Mecamylamine

Unique Feature ☆

☐ Crosses the blood-brain barrier
Unlike hexamethonium.

Current Use

☒ Smoking cessation

By reducing nicotine effects in the CNS.

[25/06/2026 11:20 AM] hssnd:  Clinical Uses

Historical Use

☐ Severe hypertension

Current Status

☒ Not used for chronic hypertension

Reason:

 Too many side effects

 Better drugs available

(ACE inhibitors, ARBs, CCBs, β -blockers, diuretics)

Limited Modern Uses

Trimethaphan

 Aortic dissection

 Hypertensive emergencies

Mecamylamine

☒ Smoking cessation

 Adverse Effects

Since all autonomic ganglia are blocked, effects occur throughout the body.

 Orthostatic Hypotension ☆☆☆

Most Important Toxicity

Normally:

Standing Up



Sympathetic Reflex



Maintains BP

With ganglion blockers:

Standing Up



No Sympathetic Reflex



Sudden BP Drop



Dizziness/Syncope

☐ Severe postural hypotension

2 Dry Mouth (Xerostomia)

Parasympathetic Block



↓ Saliva Production



Dry mouth

3 Constipation

Parasympathetic Block



↓ GI Motility



Constipation

4 Blurred Vision

Parasympathetic Block



Cycloplegia



Difficulty focusing



Blurred vision

5 Urinary Retention

Parasympathetic Block



↓ Bladder Contraction



Difficulty urinating

6 Sexual Dysfunction

Especially in males.

Autonomic Dysfunction



Impaired Erection/Ejaculation



Impotence

7 Salt & Water Retention

↓ Blood Pressure



Compensatory Mechanisms



RAAS Activation



Na⁺ + Water Retention



Edema may develop